

Emily Tumacder

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EDUCATION

Purdue University

Aug 2023 - May 2027

B.S Computer Engineering, Minor in Statistics — Honors Goss Scholar

- GPA: 3.76/4.00
- *Specialization:* Computer Systems

EXPERIENCE

Hardware Engineering Intern

May 2026 - Aug 2026

Stryker-Vocera, San Jose CA

- Prototyped visual and haptic feedback components for Vocera SyncBadge
- Tested BLE beaconing compatibility with SyncBadge and created a demo implementation for hospital use
- Developed hardware and programmed a GUI for power consumption data collection to be used during validation of PCB function

Electrical Engineering Intern

May 2025 - May 2026

Kimberly Clark, Paris TX

- Designed electrical packages for upgrading machinery and IT systems, including PLC programs, safety circuits, and routing
- Programmed a PyGame-based training simulation for line operators, focused on scalability using multithreading and OOP principles
- Used Splunk (SPL), VSphere, and OTBase to troubleshoot 100+ network switch statuses during capital projects and built documentation for plant wide network devices

Digital Systems Teaching Assistant

Jan 2026 - May 2026

Purdue University, West Lafayette IN

- Facilitated three lab sections with 30+ students each
- Debugged SystemVerilog code during design labs
- Diagnosed wiring issues in student breadboard circuits during hardware labs

RESEARCH & PROJECTS

End-to-End Aphasia Severity Prediction

Jan 2024 - Dec 2025

Advised by Dr. Yan Cong, CLA @ Purdue University

<https://doi.org/10.3389/frai.2026.1857106>

- Demonstrated the use of LLMs in automatic aphasia detection on clinical conversations through the finetuning and prompt engineering of a Llama3.2 model
- Extracted and filtered examples of speech from a database of 50k+ annotated lines of patient conversational data using Python (pandas)

BiBi: Benchtop Buddy

Dec 2025

ECE362: Intro to Microelectronics

- Designed the hardware for an autonomous desktop cleaning robot using KiCAD and simulations via SPICE

SKILLS

Programming Tools

Python, R, SystemVerilog, C, SQL, HTML/CSS, MATLAB
AutoCAD, KiCAD, ControlLogix, Splunk, TINA-TI SPICE, DMM, Oscilloscope,
Spectrum analyzer, Function generator